

Customer Churn Survival Analysis

Kaplan–Meier Estimator and Cox Proportional Hazards Model

1. Introduction

Customer churn is one of the most important business problems in subscription-based industries such as telecommunications, software-as-a-service (SaaS), banking, and streaming platforms. Understanding how long customers remain subscribed and identifying the factors associated with customer attrition are essential for improving retention strategies and maximising long-term revenue.

This project applies survival analysis techniques to customer churn data from a telecommunications company. Survival analysis is particularly appropriate in this context because it models the time until an event occurs, where the event of interest is customer churn.

Unlike traditional classification approaches that only predict whether a customer will churn, survival analysis incorporates the temporal dimension by analysing:

- how churn risk evolves over time,
- how long customers typically remain subscribed,
- and which factors accelerate or reduce churn risk.

Two classical survival analysis methods were used:

1. **Kaplan–Meier Estimator**

A non-parametric estimator used to estimate survival probabilities over time.

2. **Cox Proportional Hazards Model**

A semi-parametric regression model used to quantify the impact of multiple variables on churn risk.

2. Objectives

The objectives of this analysis were:

- Estimate the probability that customers remain subscribed over time.
- Estimate the typical customer lifetime.
- Identify variables associated with increased or reduced churn risk.
- Compare survival behaviour across contract types.
- Evaluate the proportional hazards assumption.

3. Dataset Description

The analysis used the **Telco Customer Churn Dataset**, publicly available on Kaggle.

Each observation corresponds to one customer and includes demographic, billing, and subscription-related variables.

The main variables used include:

Variable	Description
tenure	Number of months customer remained subscribed
Churn	Whether the customer left the company
MonthlyCharges	Monthly subscription fee
TotalCharges	Total amount paid by customer
Contract	Contract type
InternetService	Internet service category
PaymentMethod	Payment method
SeniorCitizen	Whether customer is a senior citizen

4. Data Cleaning and Preprocessing

Several preprocessing steps were applied before statistical modelling.

4.1 Column Standardisation

Leading and trailing spaces were removed from column names to ensure consistency during analysis.

4.2 Numeric Conversion

The variable `TotalCharges` contained non-numeric values. These entries were converted into numeric format while invalid observations were treated as missing values.

4.3 Missing Values

Observations containing missing values were removed to ensure valid model estimation.

5. Survival Analysis Framework

Survival analysis requires two fundamental variables:

5.1 Duration Variable

Customer tenure, measured in months, was used as the duration variable.

5.2 Event Variable

The event of interest was customer churn.

Customers who left the company were coded as:

$$event = 1$$

Customers who remained subscribed at the end of the observation period were treated as censored observations:

$$event = 0$$

Censoring occurs when the exact event time is unknown, but it is known that the event had not occurred during the observation period.

6. Kaplan–Meier Survival Analysis

6.1 Kaplan–Meier Survival Function

The survival function represents the probability that a customer remains subscribed beyond time t .

$$S(t) = P(T > t)$$

where:

- $S(t)$ = survival probability at time t
- T = time-to-event variable

6.2 Kaplan–Meier Estimator

The Kaplan–Meier estimator is defined as:

$$\hat{S}(t) = \prod_{t_i \leq t} \left(1 - \frac{d_i}{n_i} \right)$$

where:

- d_i = number of churn events at time t_i
- n_i = number of customers at risk immediately before time t_i

The estimator computes survival probabilities sequentially across observed event times while properly accounting for censored observations.

6.3 Overall Survival Curve

The Kaplan–Meier survival curve demonstrated a gradual decline in customer retention over time.

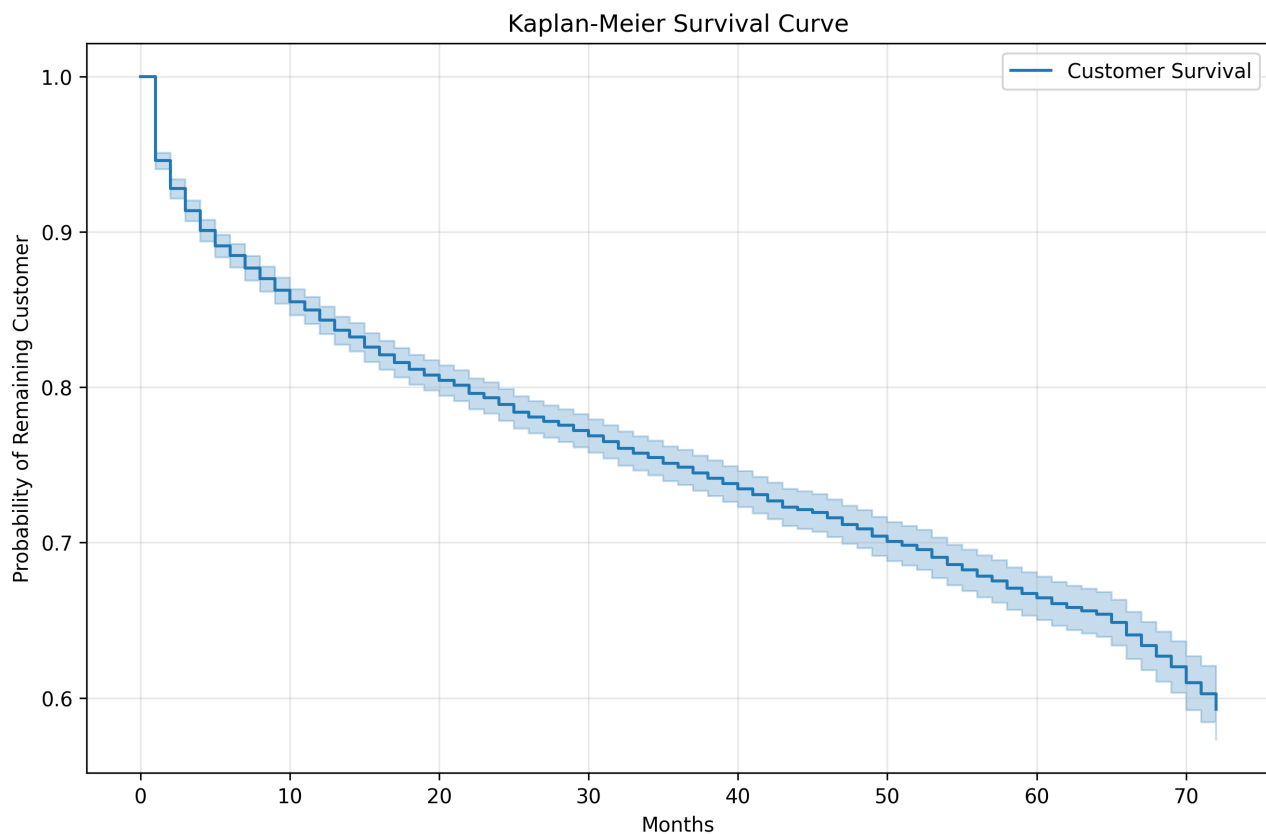
Interpretation

The curve indicates that:

- customer retention decreases progressively over time,
- churn occurs continuously throughout the customer lifecycle,
- and the steepest decline occurs during the early subscription period.

The confidence intervals remained relatively narrow, indicating stable estimates due to the large sample size.

Figure 1 — Kaplan–Meier Survival Curve



6.4 Survival Probability After 12 Months

The estimated probability that a customer remains subscribed after 12 months was:

$$S(12) = 0.8432$$

which corresponds to:

84.32 %

Interpretation

Approximately 84% of customers remained subscribed after one year, while roughly 16% churned within the first 12 months.

This suggests relatively strong short-term customer retention.

6.5 Median Customer Lifetime

The estimated median survival time was:

Median Survival Time = ∞

Interpretation

The survival curve never dropped below 50%.

This indicates that more than half of customers remained subscribed throughout the observation period, implying that the true median customer lifetime exceeds the maximum observed follow-up duration.

This result suggests relatively strong long-term retention.

7. Stratified Survival Analysis by Contract Type

Separate Kaplan–Meier survival curves were estimated for each contract category:

- Month-to-month
- One-year contracts
- Two-year contracts

7.1 Results

The survival curves reveal substantial differences between contract types.

Month-to-Month Contracts

Month-to-month customers exhibited the steepest decline in survival probability, indicating substantially higher churn rates.

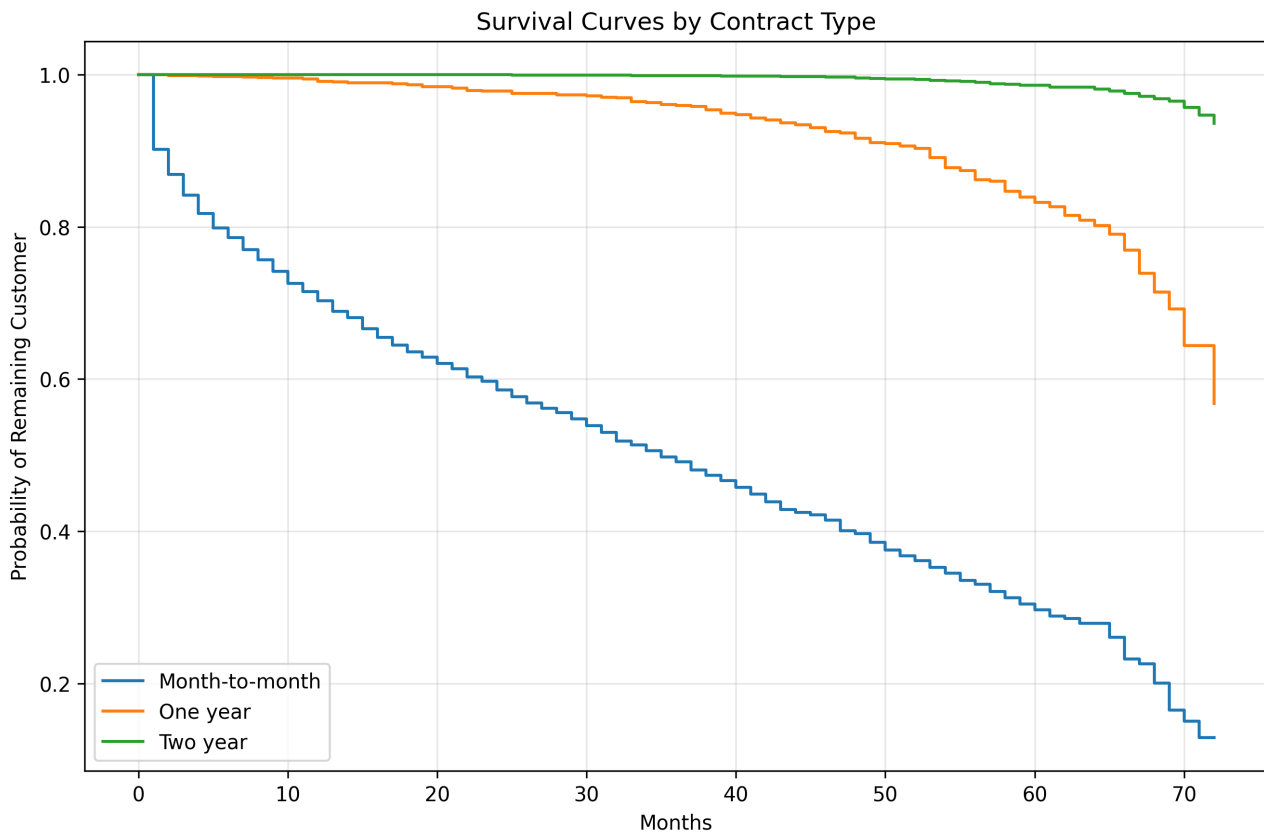
One-Year Contracts

Customers with one-year contracts demonstrated considerably better retention behaviour.

Two-Year Contracts

Two-year contracts exhibited the strongest retention performance, with survival probabilities remaining close to 1 throughout most of the observation period.

Figure 2 — Survival Curves by Contract Type



7.2 Business Interpretation

These findings strongly suggest that contract duration is one of the most important determinants of customer retention.

Longer contracts likely reduce churn because:

- customers are contractually committed,
- switching costs increase,
- and long-term customers may develop stronger service loyalty.

8. Cox Proportional Hazards Model

8.1 Cox Proportional Hazards Equation

The Cox proportional hazards model estimates how predictor variables influence churn risk over time.

The hazard function is defined as:

$$h(t | X) = h_0(t)e^{\beta X}$$

where:

- $h(t|X)$ = hazard function conditional on predictors
- $h_0(t)$ = baseline hazard function
- X = vector of predictor variables
- β = regression coefficients

The Cox model is semi-parametric because it does not require specification of the baseline hazard function.

8.2 Hazard Ratio Equation

Hazard ratios are computed as:

$$HR = e^{\beta}$$

Interpretation:

- $HR > 1$ indicates increased churn risk
- $HR < 1$ indicates reduced churn risk

9. Cox Regression Results

Several variables were statistically significant predictors of churn.

9.1 Monthly Charges

Estimated hazard ratio:

$$HR = 1.037$$

Interpretation

Each unit increase in monthly charges increased churn hazard by approximately 3.7%.

This suggests that customers are sensitive to pricing increases.

9.2 Total Charges

Estimated hazard ratio:

$$HR = 0.998$$

Interpretation

Higher accumulated spending slightly reduced churn risk.

Long-term customers appear more stable and loyal.

9.3 Contract Type

One-Year Contracts

$$HR = 0.259$$

This corresponds to approximately:

74 %

lower churn risk relative to month-to-month customers.

Two-Year Contracts

$$HR = 0.022$$

This corresponds to approximately:

98 %

lower churn risk relative to month-to-month customers.

This was the strongest protective factor identified in the analysis.

9.4 Internet Service

Fiber Optic Service

$$HR = 1.994$$

Fiber optic customers exhibited nearly double the churn risk compared to the baseline category.

Possible explanations include:

- higher service costs,
- stronger customer expectations,
- or increased competition in premium service segments.

No Internet Service

$$HR = 0.248$$

Customers without internet service exhibited substantially lower churn risk.

9.5 Payment Method

Electronic Check

$$HR = 1.540$$

Electronic check users exhibited approximately 54% higher churn risk.

Mailed Check

$$HR = 1.687$$

Mailed check users exhibited approximately 69% higher churn risk.

These payment methods may reflect less digitally engaged customer groups.

10. Proportional Hazards Assumption Diagnostics

The proportional hazards assumption was evaluated using formal statistical diagnostics.

Several variables demonstrated statistically significant deviations from the proportional hazards assumption, including:

- `MonthlyCharges`
- `TotalCharges`
- contract variables,
- `InternetService_No`

The proportional hazards assumption implies that predictor effects remain constant over time.

Violations suggest that:

- some effects may vary during the customer lifecycle,
- nonlinear relationships may exist,
- or interaction effects with time may be present.

Although these violations do not invalidate the analysis, they suggest that more advanced survival modelling approaches could improve predictive performance.

Potential improvements include:

- stratified Cox models,
- time-varying covariates,
- nonlinear transformations,
- and interaction terms.

11. Discussion

This analysis demonstrates that survival analysis provides valuable insights into customer retention dynamics.

Several important findings emerged:

- Contract duration is the strongest determinant of customer retention.
- Month-to-month customers exhibit the highest churn risk.
- Higher monthly costs increase churn likelihood.
- Long-term customers become increasingly stable over time.
- Certain payment methods and service categories are associated with elevated churn risk.

The Kaplan–Meier estimator provides an intuitive visualisation of customer retention, while the Cox model quantifies the impact of multiple predictors simultaneously.

12. Limitations

Several limitations should be acknowledged:

- The proportional hazards assumption was partially violated for several predictors.
- Additional customer behavioural variables could improve predictive performance.
- External factors influencing churn were not available in the dataset.
- The analysis assumes independent censoring.

13. Conclusion

This project demonstrated the application of survival analysis techniques to customer churn modelling.

The Kaplan–Meier estimator successfully quantified customer retention probabilities over time, while the Cox proportional hazards model identified the major factors influencing churn risk.

The analysis demonstrated that:

- longer contracts substantially improve retention,
- higher monthly charges increase churn probability,
- and customer retention dynamics evolve over time.

Overall, survival analysis provides a statistically rigorous framework for understanding customer lifecycle behaviour and supporting data-driven retention strategies.